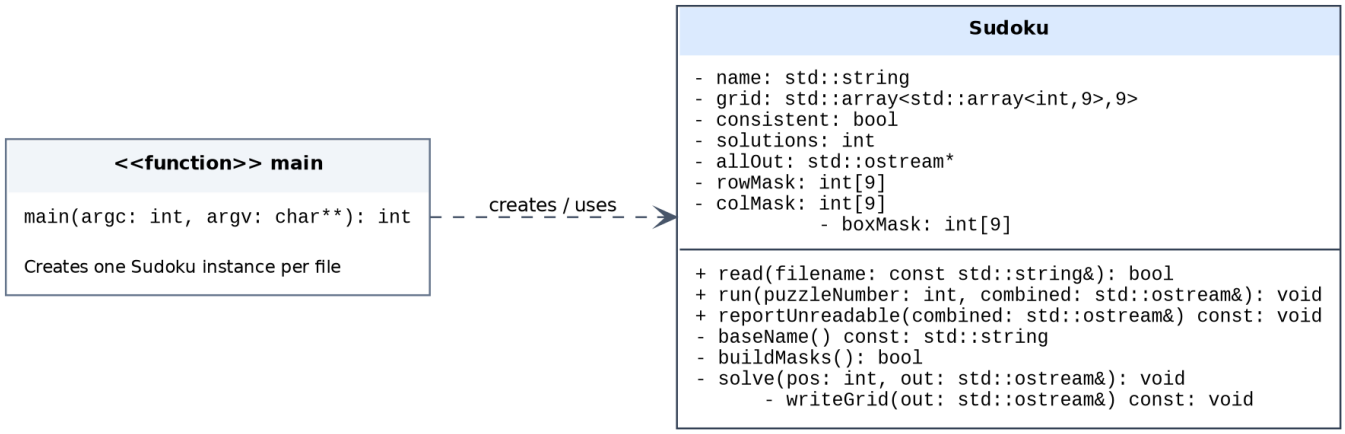
Assignment 6 | UML Class Diagram

Source: **claude.cpp** • One user-defined class: **Sudoku**



Class relationships

main() → **Sudoku:** main() is a free function, not a class. It creates a separate Sudoku object for each puzzle and calls its public methods. The dashed arrow represents this creation/use dependency.

Sudoku → **standard library types:** Sudoku stores a pointer to std::ostream, accepts std::ostream references as arguments, and uses std::string and std::array. These are existing standard-library types, not additional user-defined classes. Therefore there is no inheritance, aggregation, or association between multiple custom classes to draw.

UML visibility and signatures

The **+** symbol means public, and – means private. Every member is shown with the visibility, name, and C++ type. Each method includes its parameter list and return type; **const** is shown for const member methods.

Code-specific note

The class has no explicitly declared constructor. The single class diagram includes all 8 stored attributes and all 7 member methods found in the uploaded Claude implementation. The standalone main() function is displayed separately to show how the class is used.